CHAPTER

34

Magnification in Restorative Dentistry

The human eye may not be able to achieve the requisite precision required for restorative procedures. Visualizing the intricacies of tooth preparation always pose a challenge to the clinician. The unaided human eye can visualize only up to 0.2 mm or 200 microns. When two lines drawn parallel to each other with a distance of more than 0.2 mm, the human eye can detect it as two separate lines; however, if the separation is less than 0.2 mm, the eye will detect it as a single line.

As the age advances, the visual acuity diminishes due to presbyopia. Magnification devices significantly improve visual acuity independent of age and normal vision of any individual.

Since long, higher magnifications have been used to study cell growth, division, mutation, etc. When working without magnification, head is to be forwarded at an angle, which may cause muscle fatigue leading to a disease entity called 'tension neck syndrome (TNS)'. Such patients complain of headache and pain in interscapular muscles leading to cervical disc degeneration or spondylosis. Every dental professional is at a risk of developing musculoskeletal injury. The magnifying devices can enhance working posture by maintaining set focal range (loupes) or by location of fixed binoculars (microscope) or on LCD screen (procedure scope). It is also emphasized that improper adjustments of these devices may worsen the existing pain and injury to neck muscles. Appropriate visual enhancement makes the procedures enjoyable and also reduces risk of musculoskeletal injuries.

MAGNIFYING TOOLS

In earlier days magnifying lenses have been tried but with little success. The lenses had to be hand held by the operator. The magnification provided was not sufficient and also had the limitation of spherical and chromatic aberrations.

Common Terms Used in Magnification

i. *Usable power:* Usable power is the maximum object magnification that can be used in a given clinical situation relative to the depth and size of the restorative field. Increasing the magnification decreases the depth of field and narrows its size.
ii. *Viewing angle:* Angular position of optics allowing comfortable position of the operator.
iii. *Field of view:* Area visible under magnification.
iv. *Depth of field:* The distance between the nearest and the farthest objects.
v. *Working distance:* The distance from operator's eye to the treatment field.
vi. *Convergence angle:* Aligning of two oculars pointing at the identical distance and angle to the object or the field.
vii. *Declination angle:* The angle of operator's eyes that is inclined downward toward the work area is the declination angle.

Three basic types of magnification tools, viz. loupes, procedure scope and the operating microscopes are commonly used in dental operatory. Endoscope and orascope are also magnification gadgets.

a. Loupes

Loupes, also referred to as telescope, are usually used magnification devices in dentistry. The loupes allow working with less than 25° forward movement of head leading to less muscle fatigue (Fig. 34.1).

There are three types of loupes:

- *Flat plane (single-lens) loupe:* It consists of single lens, which is fairly inexpensive. However, plastic lens used might not be always correct. This type of loupe creates stresses compromising with posture [TTL—through the lens (fixed loupe)].
- *Galilean lens (two-lens) loupe:* Two lens system is a compact system providing up to 4.5× magnification [TTL—through the lens (fixed loupe)].



Fig. 34.1: Magnifying loupe

- *Prism loupe (prism roof design) folding the path of light:* Prism loupes are advanced type of loupes providing up to 6.0× magnification. The lens can be flipped-up during a procedure. They provide a larger field of view from longer distance minimizing stress factor (Flip-up loupe).

Loupes, despite of providing the required magnification, have their own limitations. They limit to only one magnification. The loupe becomes heavier as the magnification increases. Loupes have a fixed focal length; hence, do not contribute to ergonomics. Since the focal length of the loupes is fixed and cannot be changed, the distance from the lens to the object being narrow; the operator has to bend forward towards the patient in order to see the restorative field clearly. The occupational stresses cannot be prevented. Illumination is achieved with a head gear which casts shadow over the operating field.

Advantages

- Easy manipulation; no training required
- Inexpensive

Disadvantages

- Maximum magnification is 6×.
- Accessories like beam-splitter and video camera cannot be attached to a loupe for better field magnification
- Hither magnification loupes are heavy (flip-up loupes)
- Image is not stable due to head movement

b. Procedure Scope

The procedure scope offers optimal ergonomic benefits by facilitating neutral head posture. A camera is placed above the patient's oral cavity projecting up to 20× image onto a large flat LCD screen. The operator moves the camera freely over the patient's oral cavity while visualizing at the screen.

c. Endoscope

The traditional endoscope consists of a rigid glass rod with flexible and non-flexible variants. An optical lens with a diameter of 0.9 mm provides the clinician a magnification up to 20×.

An improved version of endoscope (2.7 mm lens diameter and 3.0 cm long rod-lens) is recommended for non-surgical procedures. The rod-lens endoscope provides greater magnification and a clear view of the field as compared to the loupes.

The endoscope should not be used to retract gingival tissue while viewing a surgical field (assistant should preferably retract the gingival tissues). Hemostasis of the surgical site, is also important before using endoscope.

d. Orascope

An orascope is a fiberoptic endoscope. Fiberoptics are made of plastics; they are light weight and are flexible. The fiberoptic endoscope (orascope) is designed for intracanal visualization.

The focus and depth of field of orascope is from 0.0 mm to infinity (∞). This allows the orascope to provide imaging of apical end of the canal without actually inserting up to that level. Orascope facilitates visualization of the treatment field at various angles and distances without losing focus and depth of field.

A 0.8 mm fiberoptic sized enables the orascope to go down into canal (canal is prepared up to size 80 or 90 at the coronal half before inserting orascope). Orascope visualization may be difficult in curved canals, because of limited flexibility of the instrument. The canal should be dried before viewing with orascope.

e. Operating Microscope

The first microscope used in clinical procedures was introduced by Apotheker in 1981. Dr. Carr introduced the first ergonomically configured operating microscope for routine dental procedures. The use of dental operating microscope has progressed over the years. Dr. Carr's statement, "You cannot treat what you cannot see", has inspired a great number of dental surgeons to add microscope to their normal armamentarium. According to Dr. Carr, the microscope is simply an avenue for greater competence and that there are some procedures that can only be performed with a microscope and almost all procedures are performed more competently using a microscope. Microscopes provide us the ability to observe and evaluate at low power; subsequently, a

higher power can be used for working and inspection. A well-illuminated field visualizing any area of the oral cavity is required to perform any procedure under the microscope.

The visual information provided by the operating microscope is, in fact, not indicative of the magnification that is being employed. The actual amount of visual information is the area under the scope (the number of horizontal pixels multiplied by the number of vertical pixels). A microscope at 10× magnification provides 100 times the amount of visual information compared to the naked-eye view (Table 34.1).

The comparison of different magnification aids is tabulated in Table. 34.2. The clinical implications are summarized in Table 34.3.

ERGONOMICS

Ergonomics is derived from Greek words, '*ergon*' meaning work and '*nomoi*' means natural laws; science of refining the design of products to optimize them for human use (literally known as engineering of human factors). One of the major advantages of the use of the dental operating microscope is the ergonomics associated with its use. The operating microscope allows the operator to sit in an upright, neutral and balanced posture with minimum neck, shoulder and lower back muscle fatigue. The increased comfort allows the operator to work more efficiently for longer periods of time. Since the eyes are focused at infinity, the eye strain is minimum. The operators working with the aid of a microscope use micro-movements and light pressure.

Researchers studying human biometrics have measured the load on the neck and shoulder muscles and have found that the load doubles with every 2.0–3.0 centimeters the head shifts forwards from the ideal upward posture.

The ideal operator zones are in the 7 to 12 O'clock positions for right-handed operators, and 5 to 12 O'clock for left-handed operators. The key to proper ergonomics is to seat yourself first in the most

Table 34.1: Magnification and resolution of various magnification systems

Magnification device	*Magnification*	*Resolution (mm)*	*Resolution (microns)*
Unaided human eye	Nil	0.2	200
Low magnification loupes	2.5×	0.08	80
High magnification loupes	6×	0.05	50
Microscope (low)	6.4×	0.031	31
Microscope (med)	10×	0.02	20
Microscope (high)	20×	0.01	10

Table 34.2: Comparison of magnification aids

Loupes	*Procedure scope*	*Microscope*
Usually fixed magnification (2 to 6×)	Variable magnification (2 to 20×)	Variable magnification (2 to 20×)
Focusing by neck	Focusing by camera	Focusing by microscope
No light	Standard light	Standard light
20° to 40° forward head posture	Head posture near neutral	Head posture near neutral
Ordinary/standard equipments	Standard equipment	Micro-equipment
3-D image convergence angle	2-D image on screen	3-D image binocular vision
Less expensive	Less expensive	Expensive

Table 34.3: Comparative clinical implications of magnification aids

Magnification type	*Head position*	*Vision*	*Operator movement*	*Direction of light*
TTL (fixed) loupes	200 to 400 forward	Three-dimensional (convergence angle)	Moderate movement, depends on working range of loupes	At an angle (less than 15°)
Flip-up loupes	200 to 300 forward	Three-dimensional (convergence angle)	Moderate movement, depends on working range of loupes	At an angle (less than 15°)
Procedure scope	Neutral	Two-dimensional (on screen)	Free movement	Parallel to operator's sight
Microscope	Neutral	Three-dimensional (binocular vision)	Restricted movement	Parallel to operator's sight

comfortable and neutral position. The operator sits with his/her thighs parallel to the floor with the hips slightly elevated, feet planted firmly on the ground, shoulders erect, straight and perpendicular to the floor and arms stretched forwards with the elbows close to the body. The microscope is then adjusted to the eyes in this position without bending forward. The patient is adjusted between the hands to maintain this neutral position of the operator.

It has been reported that the most common reason for not using a microscope is the difficulty in positioning and adjusting the eyes of the operator. Mirrors are essential to examine the beveled root end preparation in surgical endodontics, although direct vision is preferred over indirect vision, if convenient. An optimal operating position with a good front surface mirror is a critical factor in the successful use of operating microscope.

The most comfortable position of the operator is the 12 O'clock position for visualization of almost all teeth (Fig. 34.2). Right quadrant of the lower arch can also be visualized in the 9 O'clock to 11 O'clock positions. Advantage of the 9 to 11 O'clock positions is that they allow direct visualization of the occlusal surfaces of the teeth in the 3rd and 4th quadrant.

Advantages of Microscope

- Detailed view of root canal intricacies enabling the operator to be more precise.
- Multiple choices of magnification with the twist of a knob, from 2× to 20×
- Solve many traditional ergonomic problems associated with posture.
- A more relaxed eye position, as the optical properties of viewing through a microscope eliminate the need to converge the eyes for focus
- Illumination without shadows due to coaxial lighting from the microscope



Fig. 34.2: Position of operator

- An apparent increase in fine motor skills due to increased visual acuity and visibility.
- Documentation benefits from attachments in the form of digital photography and video productions.
- 'Work satisfaction' in practice. Clinicians who utilize a dental microscope enjoy more during clinical procedures due to ideal working conditions and predictable treatment outcomes.

Disadvantages of Microscope

- Indirect vision with mirrors is inevitable in non-surgical endodontics. Left-to-right and right-to-left coordination is difficult initially.
- As magnification increases, the size and depth of field decreases and also the illumination.
- Four- or six-handed dentistry is mandatory. The assistant should be well trained to be able to anticipate and pass on the relevant instruments without the operator removing his/her eyes from the eyepiece.
- It slows down the operator; each procedure takes more time to perform under the operating microscope.
- The normal instruments are too bulky; hence smaller sized instruments are required.
- Cost of the microscope and accessories.
- Learning takes time.

COMPONENTS OF MICROSCOPE

The basic components of operating microscope include:

1. Magnification
   a. Eyepiece
   b. Binocular
   c. Magnification changer
   d. Objective lens
2. Illumination
3. Accessories

1. Magnification

Microscopes can provide magnification in the range of 2× to 20× and even more. Low magnifications are excellent for routine work having a wide field of view and a good depth of focus. Midrange magnifications are used for restorative procedures. In endodontics, these are known as the 'working magnification'. Higher magnifications are used for specific tasks, such as inspection of finer details, check resected root surfaces, locating extra canals, retrieving separated instruments, etc. The field of view and the depth of focus is reduced. Overall magnification is determined by the power of the eyepiece, focal length of the

binoculars, the magnification change factor and the focal length of the objective lens.

The total magnification of a microscope is represented by the following formula:

$$\text{Total magnification value} = \frac{\text{Focal length of tube}}{\text{Focal length of object length}} \times \text{Eyepiece power} \times \text{Magnification value}$$

Magnification: Points to remember

- If the focal length of the objective lens is increased, the magnification and the illumination is decreased, while the field of view is increased.
- If the focal length of the binoculars is increased, the magnification is increased and the field of view is decreased.
- If the magnification factor is increased, the magnification is increased and the field of view is decreased.
- As the magnification increases (may be the power of eyepiece), the field of view and the depth of field decreases.

a. *Eyepiece*

The eyepiece is available in powers of 6.3×, 10×, 12.5× …. and so on. They provide the desired magnification of an object along with the focal length and magnification change factors. The viewing side of each eyepiece has a rubber cup, which is turned down if the operator wears eyeglasses (Fig. 34.3).

It has adjustable diopter settings, which range from –5 to +5 and are used to adjust for accommodation (the ability to focus the lens of the eyes). The accommodation ability decreases with age. They also adjust for refractive error (the degree to which a person needs to wear corrective eye glasses).

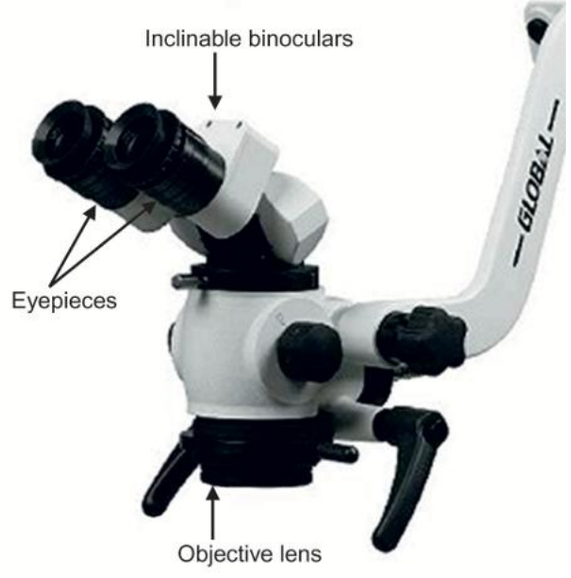


Fig. 34.3: Eyepieces and objective lens

b. *Binoculars*

The function of the binoculars is to hold the eyepieces. It allows the adjustment of interpupillary distance. They are aligned manually or with a small knob until the two divergent circles of light combine to affect a single focus. It projects an intermediate image into the focal plane of the eyepieces. As in a typical pair of field binoculars, the interpupillary distance is set by adjusting the distance between the two binocular tubes.

Binoculars come in different focal lengths. Longer the focal length of the binocular, the greater is the magnification. They are available as straight, inclined or inclinable forms.

i. *Straight binocular:* Straight binocular is oriented parallel to the optical axis of the microscope. They have the advantage of allowing the use of direct vision in both the arches.
ii. *Inclined binocular:* Inclined binocular is offset at an angle of 45°.
iii. *Inclinable binocular:* Inclinable binocular (Fig. 34.4) is adjustable for positions up to and sometimes beyond 180°. This type of binocular is most useful for endodontic procedures including surgeries. It allows the operator to look directly at the maxillary and mandibular arches. It also provides postural comfort and flexibility.

c. *Magnification Changers*

The dental microscopes have magnification changers (Fig. 34.5) to increase or decrease the magnification during restorative procedures. These are available as 3 or 5 steps manual changers or power zoom changers. Located in the head of the microscope, it consists of



Fig. 34.4: Inclinable binocular

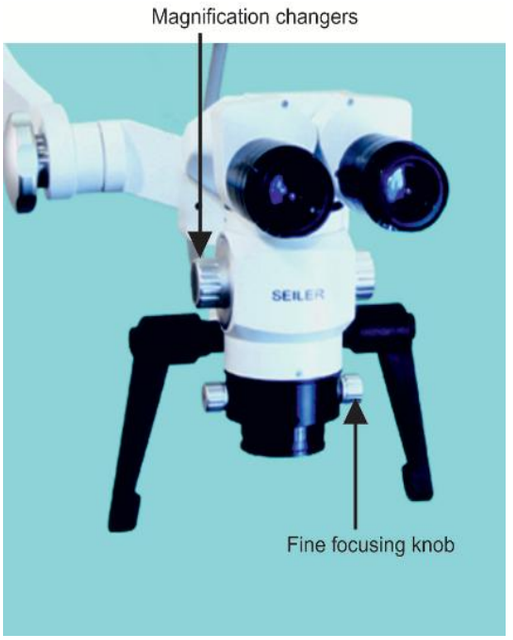


Fig. 34.5: Magnification changers

lenses that are mounted on a turret. This turret is connected to a dial that is located on the side of the microscope housing. The dial positions one lens in front of the other within the changer to produce a fixed magnification factor. Five steps changer has a second set of lenses mounted on the turret and produce five fixed powers of magnification. Rotating the dial reverses the lens position and produces a second magnification factor.

Power zoom changer is a series of lenses that move back and forth on a focusing ring to give a wide range of magnification factors. Advantage of power zoom changers is that it avoids momentary visual disruption, which is common using a 3 or 5 steps manual changer.

d. *Objective Lens*

The focal length of the objective lens determines the distance between the lens and the surgical field. A variety of objective lenses are available with focal lengths ranging from 100 to 400 mm (Table 34.4).

Table 34.4: Distance between lens and restorative field vs focal length of objective lens

Focal length of objective lens	*Distance between lens and restorative field*
100 mm	4 inches
175 mm	7 inches
200 mm	8 inches
400 mm	16 inches

A 200 mm focal length and 8 inches distance is recommended so as to have adequate room to place surgical instruments and will be close to patients.

The dental operating microscope is focused much like a laboratory microscope. The manual focusing knob (located on the side of the microscope housing), manages the distance between the microscope and the surgical field.

Before the microscope is used, it is made parfocal (the process of setting the focus for the entire range of magnification for the operator). Parfocaling is operator specific. To parfocal a microscope, a flat object is placed under the scope and focused at the highest magnification. The right eye/left eye diopter settings are unique to each person and must be set individually. Once the microscope is parfocused, accessories such as cameras and auxiliary binoculars are also adjusted accordingly.

A microscope set up should have the following features for use in endodontic applications:

- 12.5× eyepiece power
- 125 mm inclined binocular
- 200 mm objective lens
- 5-step changer, ranging from 4× to 28×

2. Illumination

The illumination in a microscope is from the light source. Three light source systems are available; 100W xenon halogen light bulb, metal halide light source and a quartz halogen bulb. The xenon light is brighter and produces light almost comparable to daylight; whereas, the halogen light produces a yellow picture.

The xenon light is better than quartz halogen, since the fiberoptic cables of halogen absorb light and have a tendency to be light deficient. Also, xenon projects a brighter and warmer light against bone and soft tissues.

The metal halide light is a compact, powerful, and efficient light source. It works under high pressure and temperature. Light is delivered via a high efficiency fiberoptic illumination, thus maintaining a cool surgical site.

The intensity of the light is controlled by a rheostat and the lamp is cooled by a fan. The light is reflected through a condensing lens to a series of prisms and then through the objective lens to the surgical field.

As the light illuminates the field, the reflected image is transmitted through the objective lens, the magnification changer lenses and the binoculars as two separate optical beams. The separation of the image beams produces the stereoscopic effect necessary to assess the depth of field.

The illumination with the operating microscope is coaxial, i.e. the light is focused between the eyepieces

in such a fashion that the clinician can look into the surgical site without any shadow. The microscope uses Galilean optics, which forms at infinity and send parallel beam of light to each eye. The eye fatigue is minimum. Loupes, on the other hand, rely on convergent vision that essentially requires an overlap of two images. This form of magnification leads to eye fatigue soon (Fig. 34.6).

In order to document the surgical proceedings, the reflected light after splitting made available to camera or auxiliary observation tube. This is known as 'beam splitter' (Fig. 34.7). The beam splitter when inserted in the optical pathway of the microscope splits the light reflected from the surgical field and directs the image to a camera and/or an auxiliary observation tube. Depending on the split ratio, most of the light is available for the operator. A traditional 50/50 beam splitter contains two cube prisms. These prisms reflect 50% of the available light across the area of each prism through partially reflective coatings. The reflected portion of the light then goes through an iris diaphragm to a video camera. The other 50% is transmitted across the area of the cube prism to the primary user (Fig. 34.8).

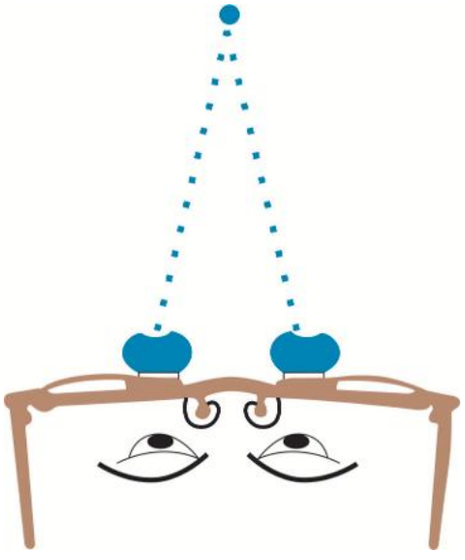


Fig. 34.6: Convergent vision leading to eye fatigue in loupes



Fig. 34.7: Beam splitter

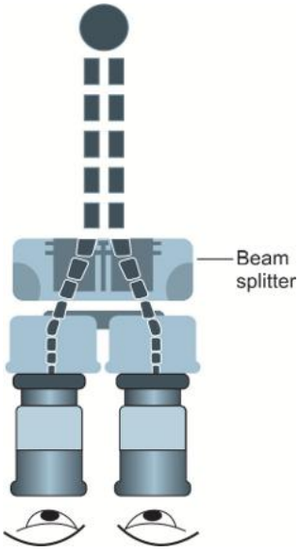


Fig. 34.8: Parallel vision created by beam splitter minimizing eye fatigue

3. Accessories

Many accessories are provided to facilitate movement during restorative procedures.

- Pistol grip or bicycle style handles can be attached to the bottom of the microscope to facilitate movement of the microscope during restorative procedures. Either of the two types is required for maneuverability.
- Observer tubes (auxiliary monocular or binocular) for assistants to see what the operator performs so that they can assist at the same magnification. LCD screen can also be provided for assistants.
- Camera and video adapters (Fig. 34.9) are used to attach video and digital cameras to the beam splitters for documentation of photographs and videos. These adapters should have the same focal length as the eyepiece so that the camera can record an image with the same magnification and field of view as that seen by the operator.



Fig. 34.9: Camera and video adapter

Use of microscope: Common errors

- Abnormal magnification: High magnification leads to narrow field of vision; low magnification may not serve the purpose. 8–10× may be adequate for diagnosis; whereas, 15–20× is ideal for restorative purposes.
- Inadequate co-ordination between operator's eye and hands, especially during movement, may lead to errors in functioning.
- Lack of knowledge and improper training may lead to error.
- Efficient assistant is also mandatory to minimize errors.

USES OF THE OPERATING MICROSCOPE IN RESTORATIVE DENTISTRY

1. Diagnosis

a. *Diagnosis of Caries*

Magnification is essential for diagnosing incipient dental caries (Fig. 34.10 A, B). The initial color alteration/demineralization on the tooth surface can only be diagnosed by microscope. High magnification makes it possible to detect hidden caries (microscopic signs of decay inside the crown).

However, ICDAS criteria of visual caries detection do not prefer magnification (use of magnification may involve unnecessary and premature invasive treatment).

b. *Soft Tissue Evaluation*

Soft tissue evaluation, in routine, may not require magnification. However, at times, inconspicuous sinus tract can be located using higher magnification.

c. *Identification of Cracks*

The superficial fracture lines in the enamel are relatively undetectable with normal vision; however, when viewed under high power, definite cracks can be diagnosed (Fig. 34.11).

2. Tooth Preparation

a. ***Cavity preparation:*** The cavity preparation using microscopes is precise; the requisite depth, taper of walls, bevels etc. can be evaluated and modified as per need (Fig. 34.12A, D).

b. ***Crown preparation:*** The margin placement and outline for crown or veneer preparation can be perfected with thorough precision under the microscope (Figs 34.13 and 34.14).

c. ***Adhesive dentistry:*** Adhesive dentistry requires ideal margins and thorough removal of infected caries. Microscopes help removal of caries close to pulp, distinguishing between infected and affected dentin.

d. ***Indirect restorations:*** The indirect restorations (metal, ceramic and composite) are examined under microscope, especially the inner surface. The irregularities, if, any, can be removed prior to seating the restoration.

e. ***Evaluating the impression:*** The impression surface is examined for any imperfections, marginal ditching, distortion, etc. under microscope; facilitates laboratory procedures.

f. ***Post-restorative evaluation:*** The restorations, over a period of time, may get deteriorated and need repair/modification/polishing, etc. Microscopes are boon for such early evaluations, enabling the operator to manage the defect and restore accordingly.

3. Preventive Therapy

Enamel and other areas are better evaluated for pit and fissure sealants, resin infiltrations, ART restorations, etc.

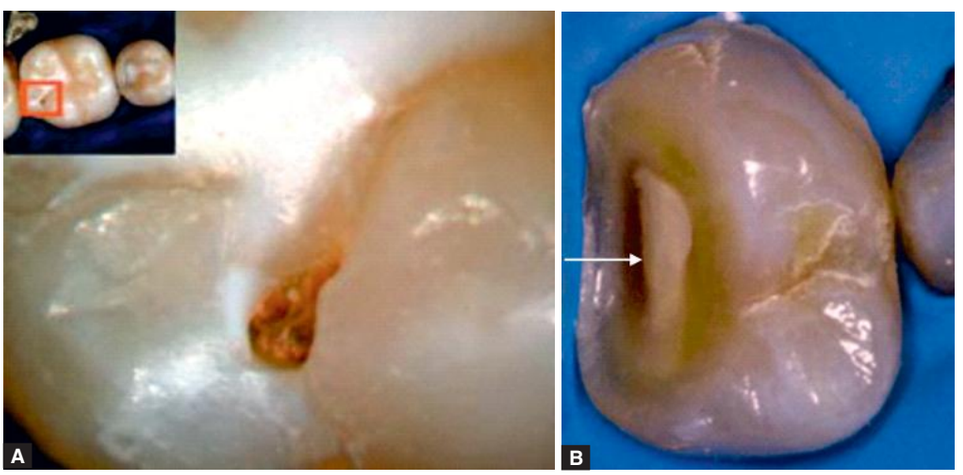
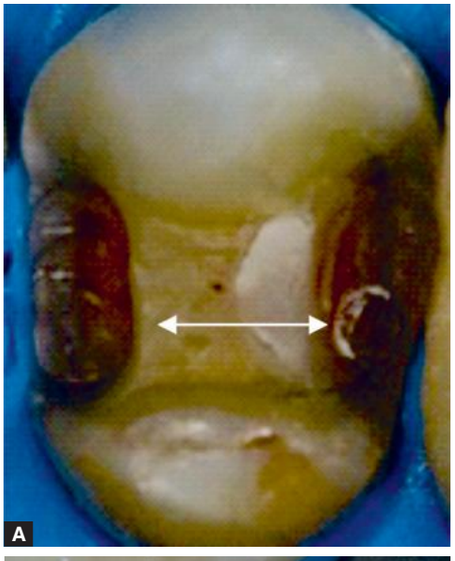


Fig. 34.10: (A) Incipient caries; **(B)** Evaluating caries at axio-cervical line angle (arrow)



Fig. 34.11: Identification of cracks



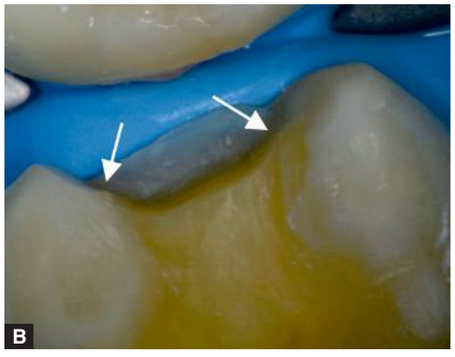


Fig. 34.12: **(A)** Evaluating axial walls (arrow); **(B)** Evaluating proximal walls in Class II cavity (arrows)

4. Documentation and Patient Education

The endodontic procedures can be videotaped using camera provided with microscope. The photographic evidence of the procedure, not only help in legal implications, but also in patient education. The clinical



Fig. 34.12C: Occlusal cavosurface without bevel (butt joint)



Fig. 34.12D: Occlusal cavosurface with bevel

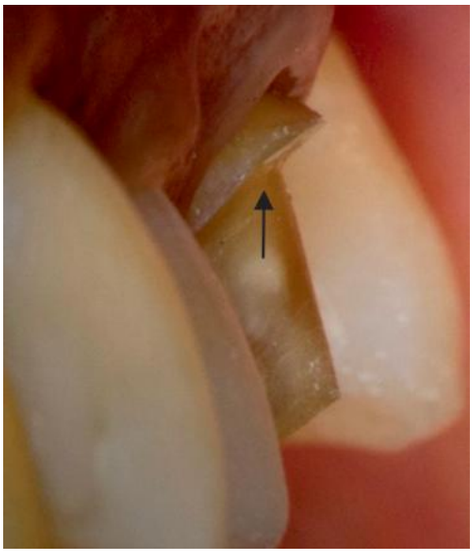


Fig. 34.13: Evaluating shoulder on labial wall (arrow)



Fig. 34.14: Evaluating margins (finish lines)

and radiographic features can be shown to the patient, in case some controversy arises, or for motivating the patient and family members.

Documentation is an important benefit of using the dental operating microscope. In this era of evidence-based dentistry, cameras attached to the microscope help in recording and documenting the treatment procedures.

It serves three purposes:

i. Educate the patients as well as the students.
ii. Better mode of communication with the referral dental surgeon (specialist).
iii. Helps maintain legal documentation of the diagnosis and the subsequent treatment.

Improved documentation has become an effective tool in practice and patient management, enabling dental surgeons to educate, counsel and convince the patients better with visuals, conforming to the proverb 'a picture speaks a thousand words'!

5. Marketing and Practice Management

The photography with microscope can be utilized as a marketing tool to impress patients and others regarding dental procedures being carried out in the clinic. These features definitely effect the growth of the practice.

ADVANCEMENTS IN MAGNIFICATION

The main difference in the use of a microscope in dentistry as compared to other fields of medicine is the different planes and angulations at which a dental surgeon needs to work within the oral cavity. Manufacturers of microscope have constantly tried to incorporate various modifications, such as inclinable binoculars, mechanical optical rotating assembly (MORA) interface, variable focal length, so as to enable minimal changes in the position and angulation of the scope during the treatment procedure. The newer models enable the operator to be seated at 12 O'clock position, providing horizontal working distance, conducive to operator's head and the patient's target object.

Zeiss OPMI PROErgo model includes feature of foot control adjustment of focal length; creates less disturbance and provides optimal ergonomics, even working for longer durations.

Varioscope, another model, is a miniature head-mounted operating microscope, provide mobility to the operator's head, in contrary to the conventional microscope, which lack such maneuverability.

Recent additions, such as fluorescent mode in the new model Extaro 300 from Carl Zeiss enables accurate visualization of demineralized zones on the enamel surface and distinguish between natural tooth structure and composite resin during its removal. Further, the no-glare mode enables the clinician to view the proper shade of the tooth and also visualizes the operating field without the light reflection from the tooth surface. The true-light mode does not necessitate the use of orange filter during curing of composite resin, prevents premature polymerization and helps identify relevant dental tissues in a natural, white-light setting. Recent fluorescent tools (Infrared 800), allows operator to see even vascular circulation.

The use of dental operating microscope in recent years has encompassed many disciplines of dentistry from endodontics to periodontics, restorative, cosmetic and esthetic dentistry. The advantage of magnification and ergonomics combined with illumination has provided a platform for increased patient care with operator's capabilities of increased work hours due to unstrained postural positions.

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